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Earth and Planetary Science Letters 248 (2006) 533–544
EPSL
www.elsevier.com/locate/epsl

Paleomagnetic constraints on the extent of the stable body of the South China Block since the Cretaceous: New data from the Yuanma Basin, China

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Received 20 February 2006; received in revised form 15 June 2006; accepted 15 June 2006
Available online 14 July 2006
Editor: R.D. van der Hilst

Abstract

We undertook a paleomagnetic investigation of Cretaceous red sandstones at 30 sites within the Yuanma Basin in western Hunan, central South China Block (SCB), China. The aim of this study is to further constrain the extent of the SCB that has been stable since the Cretaceous. We isolated the characteristic directions of higher temperature components (HTCs) with an unblocking temperature of ~650 to 690 °C by progressive thermal demagnetization and principal component analysis. The optimal concentrations of global mean HTC directions calculated using the direction–correction tilt test were achieved at 92.7±59.1%, 87.9±25.0%, and 93.3±19.7% untilting indistinguishable from 100% untilting, for Early Cretaceous (K1, 13 sites), Late Cretaceous (K2, 17 sites) and K site groupings (K1+K2, 30 sites), respectively. This treatment yielded positive tilt tests, and the HTCs are considered to be primary magnetic components acquired before tilting of the sedimentary layers. Early (K1) and Late (K2) Cretaceous paleomagnetic poles calculated using the site mean directions untilted completely are located at similar positions; this suggests the absence of local rotations linked to Cretaceous strain within the study area. The Cretaceous (K1+K2) mean paleomagnetic pole calculated using virtual geomagnetic poles from all sites is in agreement with previously reported Cretaceous paleomagnetic poles from the eastern and western SCB. This indicates that the Hunan region has been part of the stable body of the SCB since the Cretaceous and that the tectonic influence of the India–Asia collision did not extend to within the stable body of the SCB. The reference Cretaceous pole position (latitude=80.1°N, longitude=204.1°E, A_{95}=2.5°) calculated using eight Cretaceous poles from the stable body of the SCB is similar to those determined from stable Eurasia and the North China Block (NCB). This result indicates that the India–Asia collision caused no relative motion, as detected by paleomagnetic analysis, among stable Eurasia, NCB, and the stable body of the SCB.
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Keywords: Paleomagnetic pole; South China Block (SCB); Cretaceous; Red sandstone; Tectonics

1. Introduction

The indentation of the Indian continent into the Asian continent resulted in the tectonic deformation of southeastern and eastern Asia, especially the Indochina Block. Paleomagnetic investigations [1–6] present

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doi:10.1016/j.epsl.2006.06.029

evidence that indentation has resulted in some areas of the Indochina Block, excluding the Song Da Terrane [7], being extruded to the southeast and rotated clockwise. Paleomagnetic investigations [8–11] also demonstrate that indentation caused small but differential tectonic movement of regions along the Red River Fault (RRF) within the South China Block (SCB). These paleomagnetic results support the occurrence of extrusion Tectonics associated with collision between India and Asia [12–15].

The results of paleomagnetic investigations [18–21] also indicate that most of the SCB has suffered no apparent effects from the India–Asia collision (Fig. 1). Morinaga and Liu [18] reported the paleomagnetism of Cretaceous red sandstones distributed over the Zhejiang, Fujian, and Guangdong Provinces of the eastern SCB. The authors found a good agreement among the paleomagnetic poles of these three regions and also between these poles and the Cretaceous mean pole for the Sichuan Province of the western SCB, as determined by Gilder et al. [8]. Morinaga and Liu [18] concluded that there is no relative tectonic movement between the eastern and western sides of the SCB and indicated that a large part of the SCB, excluding the 400-km-wide swath along the RRF, has behaved as a stable body since the Cretaceous. The idea that the 400-km-wide swath along the RRF within the SCB has been tectonically deformed was reported at first by Liu and Morinaga [9]. Morinaga and Liu [18] also found a good agreement between the mean Cretaceous poles of the stable body of the SCB and stable Eurasia [22].

Paleomagnetic data for the Cretaceous to Paleogene, prior to the India–Asia collision, are scarce for the central part of the proposed stable body of the SCB. To define the extent of the stable body, it is necessary to obtain additional Cretaceous paleomagnetic data from the central part of the body. Narumoto et al. [23] recently reported that Late Cretaceous red beds from the Yichang

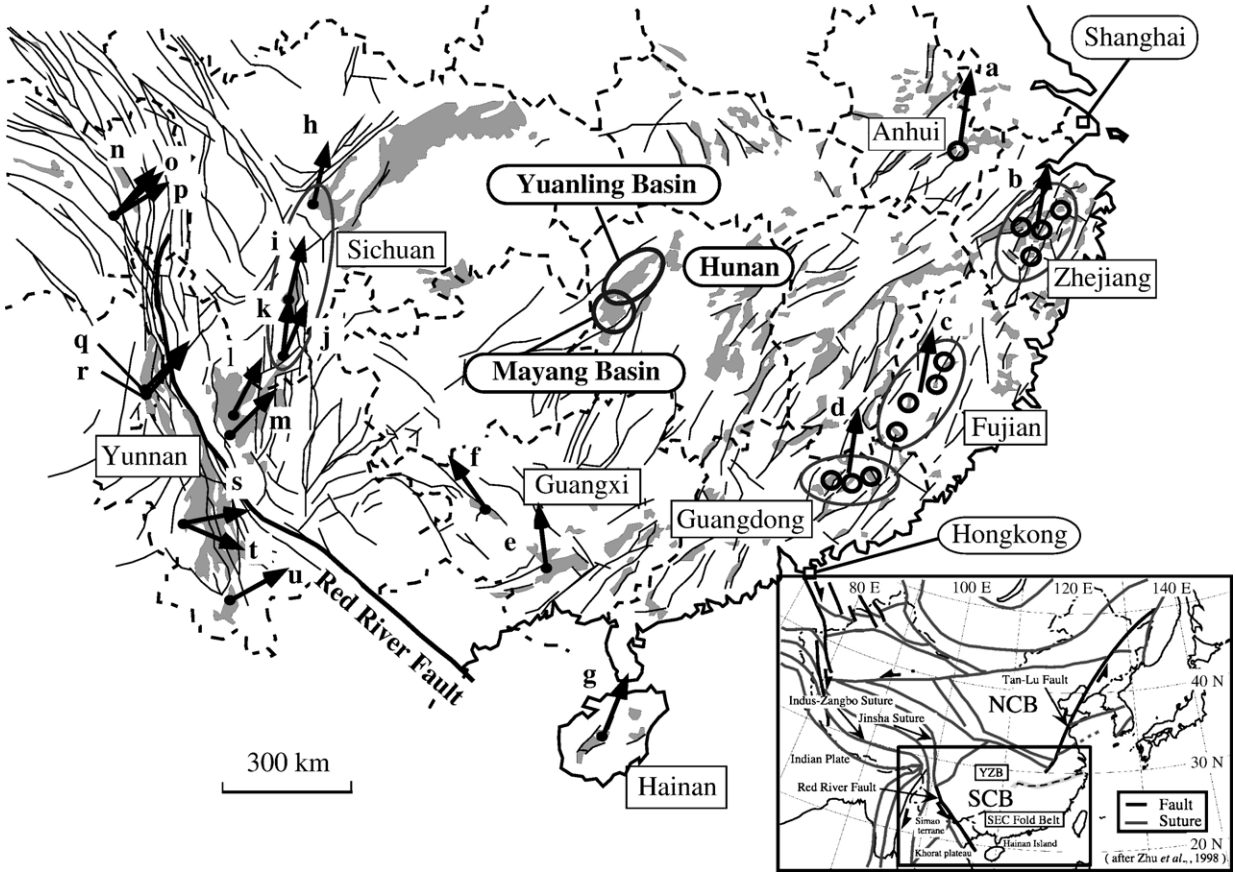


Fig. 1. Simplified geological map of the South China Block (SCB) and adjacent areas, showing the distribution of faults and Cretaceous basins [16]. Inset map is a tectonic map of East Asia, modified from Zhu et al. [17], showing the location of the main map. Arrows in the main map show the mean declinations reported in previous studies: a [19], b–d [18]; e, f [8]; g [9]; h [20]; i [21]; j, k, p, n [1]; l [10]; m [11]; o [2]; q [3]; r [4]; s, u [5]; t [6]. The two ellipses of the central part of the SCB indicate the current study, within Hunan Province.

area in the middle-northern part of the SCB record inclination values that are 26.3±6.4° shallower than expected. As the shallowing of the inclination was observed for almost all laminated samples, a useful paleomagnetic pole for the Late Cretaceous is not available [23]. In the present paper, we present new paleomagnetic results for Cretaceous red sandstones distributed throughout the Hunan Province in the central SCB, and consider whether the Hunan area is part of the stable body of the SCB. We also discuss the tectonic implications of our paleomagnetic results.

2. Geological setting, sampling, and laboratory procedures

Cretaceous red sandstones are widely distributed within the SCB. Most of these red sandstones or red beds were deposited in extensional basins that formed during Mesozoic extensional Tectonics [24]. Most of the tilting (folding) in the extensional basins is thought to have occurred contemporaneously with the structural and stratigraphic development of the basins [25]. Such a timing explains the large thickness (~1000 m) of sedimentary layers deposited in basins within the eastern SCB. The age of these basins is considered to be Cretaceous on the basis of fossil vertebrates, pelecypods, ostracods, gastropods, plants, and dinosaur eggs, although fossils are rare [26].

Sedimentary basins that contain red beds are widely distributed throughout Hunan Province. The NE-trending Yuanling and Mayang Basins are located in the western part of Hunan Province (Fig. 1), and are interpreted as Cretaceous extensional basins [27]. The combined name for these two basins is the Yuanma Basin, which has an area is about 10,000 km^2 [28]. We collected paleomagnetic core samples of red sandstone from 15 sites in the Yuanling Basin and 16 sites in the Mayang Basin (Fig. 2). Within the Yuanma Basin, we sampled from the Dongjing (7 sites; stratigraphically oldest), Lanlong (6 sites), and Shenhuangshan Formations (18 sites; stratigraphically youngest), which have thicknesses of about 560 m, 750 m, and 1630 m, respectively [29,30]. The ages of the Dongjing and Lanlong Formations are estimated to be Early Cretaceous (K1), while the Shenhuangshan Formation is estimated to be Late Cretaceous (K2), as determined from rare ostracod and pelecypod fossils [29]. As the lithology of the red beds is very similar to the Cretaceous red beds in the eastern part of the SCB, it is reasonable to accept a Cretaceous age for these units. Even if the age is incorrect, it is considered reliable that the age of the rocks is older than the India–Asia collision.

Paleomagnetic cores were sampled using a portable gasoline-powered drill and oriented using a magnetic compass. The latitude and longitude of the sites were determined using GPS (Pyxis, SONY). The present geomagnetic direction at each site was calculated using the International Geomagnetic Reference Field, 2000 [31]. Remanent magnetization of samples was measured using a Natsuhara Giken SMM-85 spinner magnetometer. Progressive thermal demagnetization was conducted to evaluate the magnetic stability of samples and eliminate viscous and unstable magnetic components, using a Natsuhara Giken TDE-91C thermal demagnetizer. The direction of characteristic remanent magnetization (ChRM) was computed from principal component analysis [32].

3. Results of magnetic measurements and interpretation

3.1. Progressive thermal demagnetization

The samples were thermally demagnetized using an average of 15 temperature steps until 680 or 690 °C. Almost all the samples showed stable behavior throughout the thermal treatment. Typical examples are shown in Fig. 3, which are the orthogonal plots [33] for K1 and K2 samples, respectively. Most of samples show a single magnetization component stable up to 650 to 690 °C. Some samples have a scanty viscous component that is

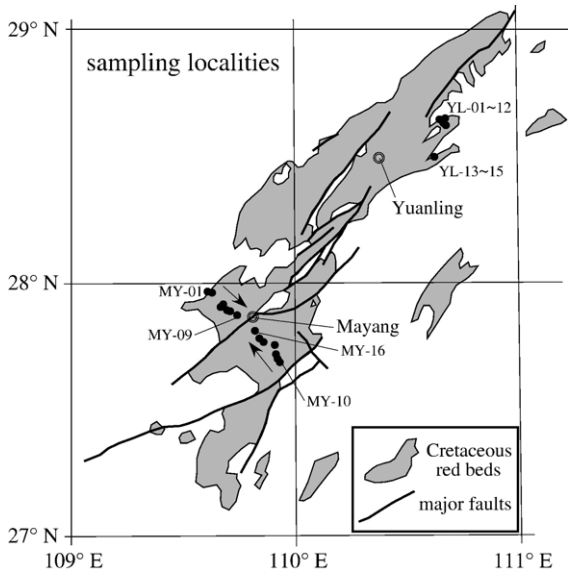
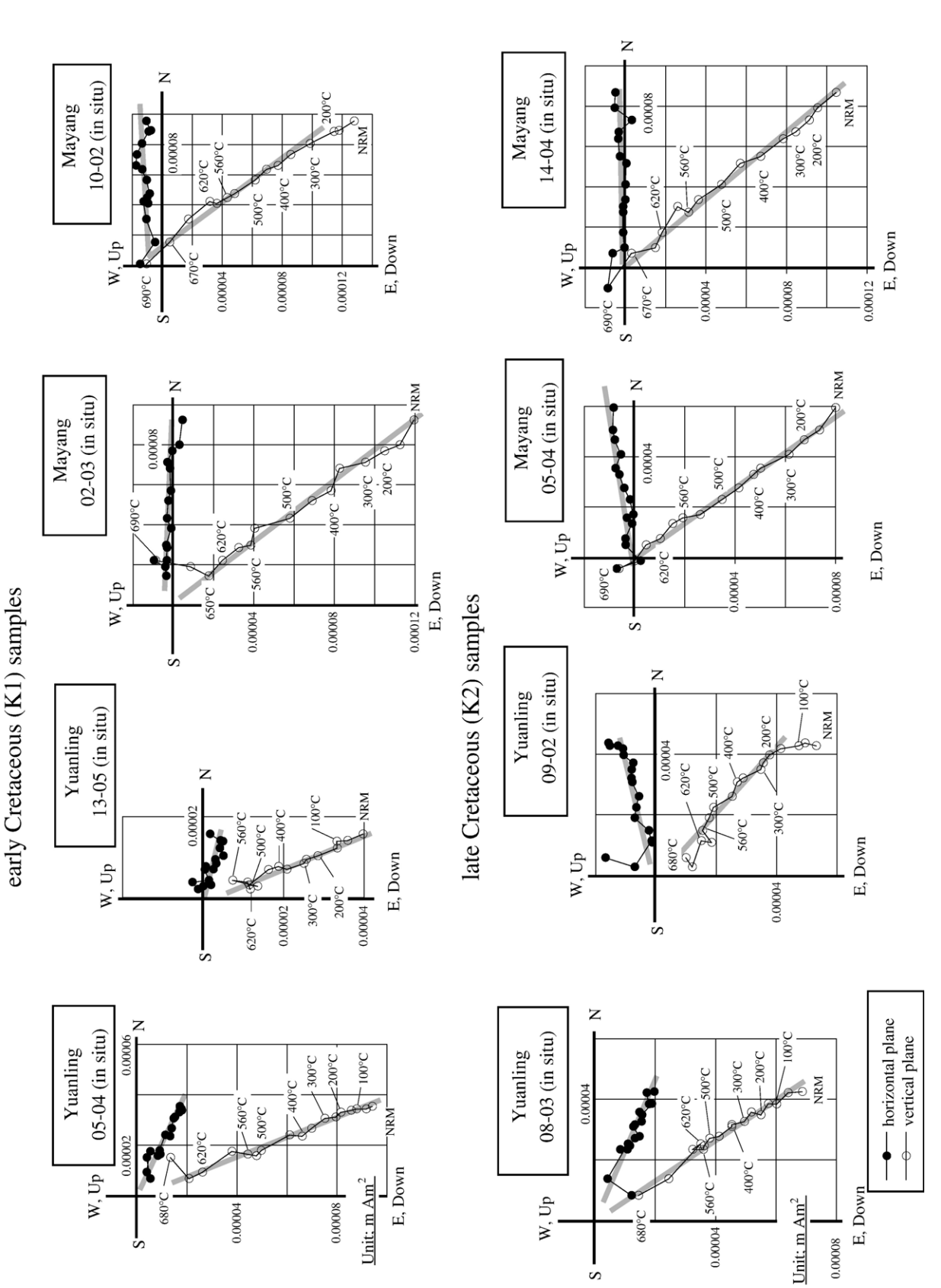


Fig. 2. Simplified geological map showing the distribution of Cretaceous red beds and major faults around the Yuanma Basin [16]. Solid circles represent the sampling localities.


early Cretaceous (K1) samples
late Cretaceous (K2) samples
Yuanling 05-04 (in situ)
Yuanling 13-05 (in situ)
Mayang 02-03 (in situ)
Mayang 10-02 (in situ)
Yuanling 08-03 (in situ)
Yuanling 09-02 (in situ)
Mayang 05-04 (in situ)
Mayang 14-04 (in situ)
W, Up
S
N
E, Down
NRM
Unit: m Am2
horizontal plane
vertical plane

eliminated by lower temperature demagnetization at about 300 °C. The single component that is stable up to higher temperature (HTC), however, generally does not trend toward the origin of the orthogonal plot. This observation is considered to be due to new magnetic grains formed in the furnace, or an apparent phenomenon reflecting the relatively low sensitivity of the spinner magnetometer as the remanence intensity of samples following higher temperature demagnetization is approximately equal to the noise level of the magnetometer. The high unblocking magnetization component of investigated samples indicates that it is carried by hematite.

3.2. Determination of paleomagnetic directions

We regard the component with an unblocking temperature in excess of ~620 °C (HTC) to be the ChRM of the samples, which is useful for paleomagnetic analysis. All the paleomagnetic results are listed in Table 1, while Figs. 4–6 show the site mean directions for the Early Cretaceous (K1), Late Cretaceous (K2), and K1+K2, respectively. *In situ* (geographical) mean directions of several K2 sites are sub-parallel to the geocentric axial dipole field (GDF) (Fig. 5). The strata at these sites, however, are more gently tilted (less than 10°) than the strata at other sites; we therefore consider that the ChRMs of these sites are not secondary magnetizations that were acquired recently or following tilting.

For the mean directions of the K1, K2, and K (K1+K2) site groupings, the optimal concentration calculated by tilting correction at a 5% step is achieved at 90, 90, and 95% untilting, respectively (Table 1). Mean directions after these incomplete untiltings are approximately equal to the mean directions achieved via complete (100%) untilting. The direction–correction (DC) tilt test [34] gives optimal concentrations at 92.7±59.1%, 87.9±25.0%, and 93.3±19.7% untilting indistinguishable from 100% untilting, for the K1, K2, and K site groupings, respectively. This treatment provides positive tilt tests for all three groupings. Thus, we interpret the ChRM components (HTCs) to be near-primary origin before tilting. All the samples show normally magnetized remanence. This is also evidence that the remanence is primary, because the Cretaceous Long Normal Superchron represents a large proportion of Cretaceous time. We therefore adopt the completely (100%) untilted ChRM directions as the Cretaceous paleomagnetic field directions.

4. Discussion and implications

Paleomagnetic poles for each sampling site were calculated using the mean directions of the completely (100%) untilted data. The geographic (Yuanling and Mayang sites) and age means (K1, K2 and K1+K2 sites) of the paleomagnetic poles are shown in Fig. 7. The poles from the Yuanling and Mayang Basins are indistinguishable from each other at the 95% confidence level. The poles for K1 and K2 are also indistinguishable from each other at the 95% confidence level. The similar positioning of these paleomagnetic poles suggests the absence of Cretaceous and post-Cretaceous tectonic movement in this area. Such an observation is also recognized, for instance, from K1 and K2 paleomagnetic poles from Zhejiang Province [18], and provides evidence that within some areas of the SCB there were no local rotations linked to Cretaceous and post-Cretaceous strain.

The Cretaceous mean pole calculated from the virtual geomagnetic poles of all sites is: latitude=82.0°N, longitude=178.6°E (k=45.9 and A_{95}=3.9°). This mean pole position agrees with previously determined Cretaceous poles from the eastern SCB (four poles from the Zhejiang, Fujian, Guangdong [18], Anhui and Shandong Provinces [19], and the western SCB (one pole from Sichuan Province, compiled by Gilder et al. [8]) at the 95% confidence level (Table 2 and Fig. 8-left). This agreement in pole positions indicates that the Hunan region has been part of the stable body of the SCB since the Cretaceous and that therefore there has no relative movement among large areas within the SCB. This further supports the hypothesis of Morinaga and Liu [18] that a large part of the SCB (excluding a 400-km-wide swath along the RRF [9]) has behaved as a stable body since the Cretaceous.

Enkin et al. [22] compiled the published Permian to present-day paleomagnetic poles for the major Chinese blocks (e.g., NCB, SCB, and Tarim Blocks) and constructed apparent polar wander paths for the major blocks after critically selecting reliable paleomagnetic poles. The authors recognized a streaked distribution of paleomagnetic poles for most time periods along a small circle centered on the sampling region and concluded that much of China has been affected by minor (<20°) differential tectonic rotations. This deformation is interpreted as being caused in part by the extrusion of the Chinese blocks away from the site of India–Asia collision. Cretaceous paleomagnetic poles from the stable body of the SCB, however, are not streaked (Fig. 8-left).

Fig. 3. Representative plots of progressive thermal demagnetization results for Early Cretaceous (K1; upper figures) and Late Cretaceous (K2; lower figures) red sandstone samples from the Yuanma Basin. The plots are orthogonal vector projections onto the horizontal plane (solid circles) and the N–S vertical plane (open circles). Numbers adjacent to the symbols indicate temperature steps (°C). *In situ* (geographic) coordinates are used.

Table 1
Paleomagnetic results for samples of Cretaceous red sandstones from the Yuanling and Mayang Basins (Yuanma Basin), Hunan Province, China

Site no.	Site		Bedding		*N**	*In situ*	
	Latitude (°N)	Longtitude (°E)	Strike (°)	Dip (°)		Declination (°)	Inclination (°)
Yuanling Basin							
YL-01	28.61	110.69	216	21.0	8	41.4	48.7
YL-02	28.61	110.69	251	22.5	12	42.6	64.2
YL-03	28.62	110.69	218	19.5	11	39.3	52.9
YL-04	28.62	110.69	243	22.5	7	28.9	53.0
YL-05	28.62	110.69	234	16.5	10	33.9	64.8
YL-06	28.62	110.69	218	19.5	9	40.6	25.0
YL-07	28.62	110.69	240	20.5	6	37.7	59.1
YL-08	28.63	110.68	219	21.0	7	25.6	57.0
YL-09,10	28.63	110.67	78	4.0	12	−4.9	42.7
YL-11	28.63	110.67	228	9.0	9	7.6	49.7
YL-12	28.63	110.67	96	25.5	11	−4.1	15.4
YL-13	28.49	110.63	232	12.0	7	27.3	67.7
YL-14	28.49	110.63	299	20.0	8	23.9	70.8
YL-15	28.49	110.63	247	31.0	11	45.0	67.8
Mean	*In situ*				14 sites	25.3	54.3
(Yuanling)	*Untitled*(100%)				14 sites		
Mayang Basin							
MY-01	27.95	109.60	237	11.0	8	−0.5	58.2
MY-02	27.94	109.63	76	9.0	6	−5.8	54.7
MY-03	27.90	109.67	312	6.0	5	8.1	53.8
MY-04	27.91	109.67	328	3.0	9	−9.4	51.2
MY-05	27.90	109.70	50	4.0	9	−5.8	50.6
MY-06	27.89	109.70	266	11.0	7	14.3	59.3
MY-07	27.88	109.71	37	17.0	6	6.0	48.5
MY-08	27.87	109.71	333	19.0	5	−0.7	65.6
MY-09	27.87	109.74	346	6.5	5	12.7	58.9
MY-10	27.68	109.92	308	9.0	8	−7.3	50.1
MY-11	27.70	109.91	294	14.0	6	14.2	60.5
MY-12	27.72	109.90	317	6.0	9	−13.8	47.4
MY-13	27.75	109.91	68	13.5	5	−0.9	44.6
MY-14	27.77	109.86	48	6.0	6	−2.1	51.2
MY-15	27.78	109.83	228	10.0	6	6.2	57.7
MY-16	27.81	109.82	260	24.0	6	53.5	64.9
Mean	*In situ*				16 sites	2.6	55.6
(Mayang)	Untitled(100%)				16 sites		
Mean (K1)	*In situ*				13 sites	26.8	58.2
	Untitled(90%)				13 sites		
	Untitled(100%)				13 sites		
Mean (K2)	*In situ*				17 sites	4.2	52.6
	Untitled(90%)				17 sites		
	Untitled(100%)				17 sites		
Mean	*In situ*				30 sites	13.1	55.5
(K1+K2)	Untitled(95%)				30 sites		
	Untitled(100%)				30 sites		

*N**; number of samples having significant components. VGP#; virtual geomagnetic pole calculated using 100% untitled data.

Table 1 (*continued*)

Untitled (100%)		*k*	α95	VGP#		Age	Formation
Declination (°)	Inclination (°)		(°)	Latitude (°N)	Longtitude (°E)		
Yuanling Basin							
18.1	46.6	76.9	6.4	74.1	199.0	K1	Dongjing
16.3	49.0	396.2	2.2	75.7	191.5	K1	Lanlong
15.1	49.2	421.1	2.2	76.8	190.7	K1	Lanlong
11.8	37.6	245.7	3.9	76.9	233.3	K1	Lanlong
9.1	55.8	665.2	1.9	79.1	153.1	K1	Lanlong
31.5	24.5	73.4	6.1	56.7	222.4	K1	Lanlong
14.4	47.4	111.7	6.4	77.4	197.6	K1	Lanlong
0.6	47.6	53.6	8.3	89.5	192.1	K2	Shenhuangshan
−4.4	46.6	265.1	2.7	86.0	10.6	K2	Shenhuangshan
0.6	43.5	135.0	4.4	86.7	281.2	K2	Shenhuangshan
−6.7	40.4	75.5	5.3	81.8	339.4	K2	Shenhuangshan
6.7	60.7	248.6	3.8	75.7	131.3	K1	Dongjing
26.1	50.8	319.6	3.1	67.3	186.5	K1	Dongjing
7.1	46.2	37.3	7.6	83.7	207.6	K1	Dongjing
		17.6	9.7			K1+K2	
10.7	46.7	49.5	5.7x	80.8	199.0	K1+K2	
(k=46.6, A_{95}=5.9°)							
Mayang Basin							
−7.7	48.6	407.2	2.7	83.1	34.8	K1	Dongjing
−3.2	63.6	252.3	4.2	72.5	102.1	K1	Dongjing
12.1	48.6	324.9	4.3	79.3	187.9	K2	Shenhyangshan
−6.0	50.0	423.8	2.5	84.0	50.0	K2	Shenhuangshan
−2.8	53.8	208.1	3.6	83.1	90.0	K2	Shenhuangshan
10.0	48.7	844.6	2.1	81.1	186.0	K2	Shenhuangshan
26.4	54.6	70.1	8.1	66.4	175.2	K2	Shenhuangshan
24.6	53.2	123.8	6.7	68.1	178.0	K2	Shenhuangshan
21.3	55.6	281.8	4.6	70.2	169.7	K2	Shenhuangshan
−0.9	43.4	390.9	2.8	87.5	308.9	K1	Dongjing
16.9	46.7	247.4	4.3	75.1	195.0	K2	Shenhuangshan
−8.6	44.2	141.1	4.3	82.1	8.7	K2	Shenhuangshan
6.1	56.9	132.2	6.7	79.0	136.1	K2	Shenhuangshan
3.4	55.6	1998.9	1.5	81.2	128.0	K2	Shenhuangshan
−3.4	50.4	349.4	3.6	85.5	69.3	K2	Shenhuangshan
24.9	48.7	391.1	3.4	68.1	189.0	K2	Shenhuangshan
		63.9	4.7			K1+K2	
6.9	52.0	75.2	4.3	82.1	157.8	K1+K2	
(k=48.3, A_{95}=5.4°)							
		26.3	8.2			K1	
13.1	49.6	43.2	6.4	(maximum k)		K1	
12.0	48.5	42.9	6.4	79.8	187.5	K1	
(k=40.4, A_{95}=6.6°)							
		32.3	6.4			K2	
6.0	50.6	81.5	4.0	(maximum k)		K2	
6.2	50.3	79.1	4.0	83.5	168.1	K2	
(k=51.3, A_{95}=5.0°)							
		25.0	5.4			K1+K2	
8.9	49.9	58.0	3.5	(maximum k)		K1+K2	
8.8	49.6	57.9	3.5	82.0	178.6	K1+K2	
(k=45.9, A_{95}=3.9°)							

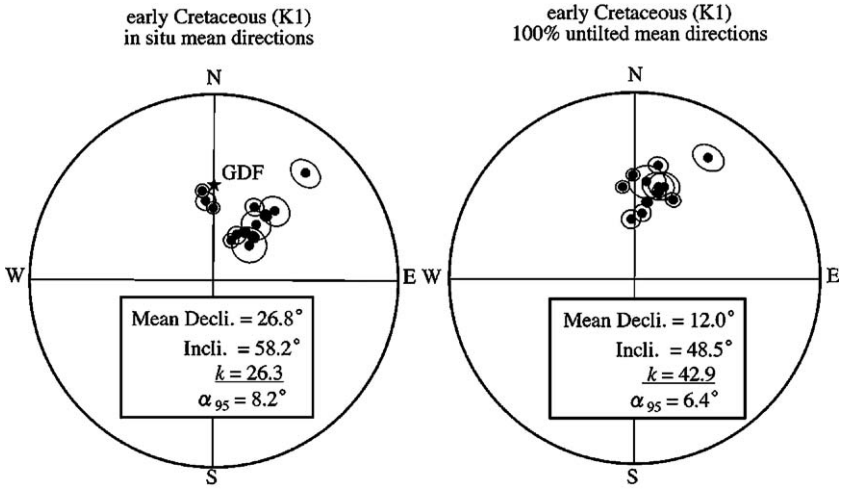


Fig. 4. Equal-area projections of the site mean directions for Early Cretaceous (K1) red sandstones from the Yuanma Basin. The plots show *in situ* (left) and 100% (right) untilted directions. ★: geocentric axial dipole field direction.

Within the SCB, the India–Asia collision resulted in local tectonic deformation only in the vicinity of the RRF; data from the present study and a previous report [18] demonstrate that the effects of the India–Asia collision do not extend to the interior of the stable body of the SCB, although it is possible that the stable body has moved as a coherent whole in response to the collision.

The global mean Cretaceous paleomagnetic pole calculated using the above-mentioned eight pole positions from the eastern, western, and central parts of the SCB is: latitude=80.1°N, longitude=204.1°E (k=478.9 and A_{95}=2.5°) (Table 2). This global mean pole is worth using as a reference Cretaceous paleomagnetic pole for the stable body of the SCB when discussing relative tectonic movements within the SCB, especially for relative movements between the SCB and other tectonic units and for areas in the vicinity of the RRF where local tectonic rotations and translations have been previously documented [8–11].

Enkin et al. [22] excluded Chinese paleomagnetic poles from the worldwide selection data [35] and defined Early Cretaceous (latitude=77.8°N, longitude=208.4°E, A_{95}=7.0°) and Late Cretaceous paleomagnetic poles (latitude=77.9°N, longitude=203.8°E, A_{95}=14.2°) for

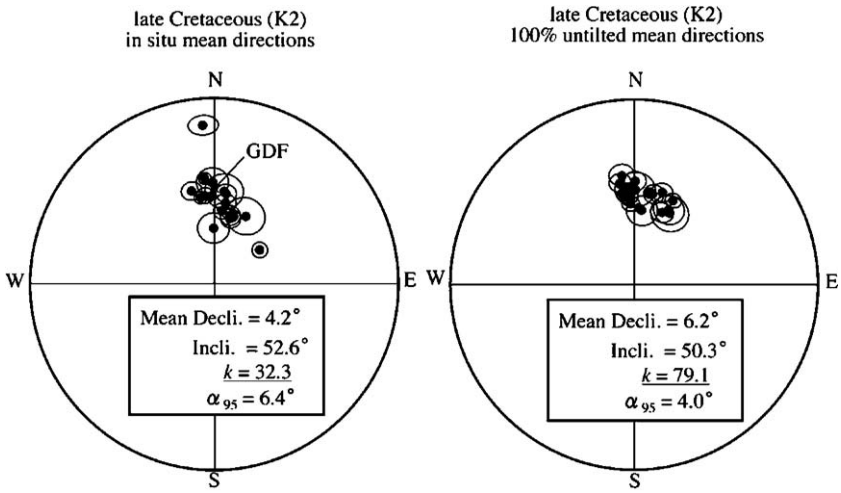


Fig. 5. Equal-area projections of the site mean directions for Late Cretaceous (K2) red sandstones from the Yuanma Basin. The plots show *in situ* (left) and 100% (right) untilted directions. ★: geocentric axial dipole field direction.

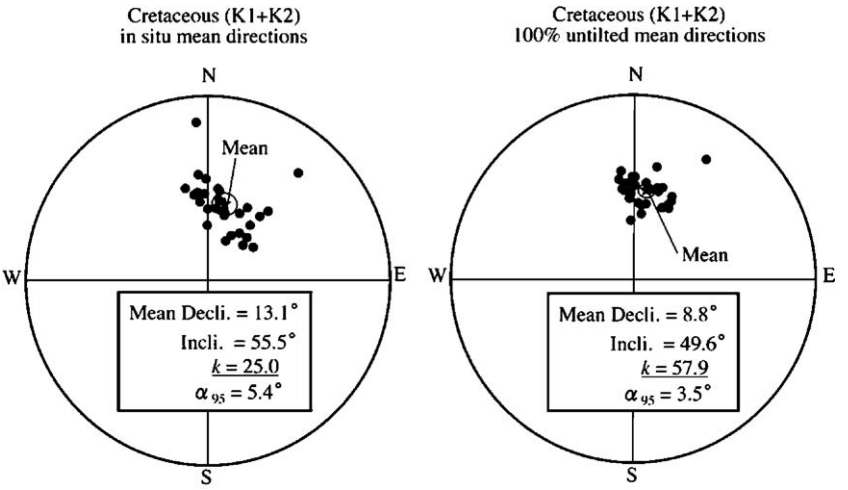


Fig. 6. Equal-area projections of the site mean directions for Cretaceous (K1+K2) red sandstones from the Yuanma Basin. The plots show *in situ* (left) and 100% (right) untilted directions. ★: geocentric axial dipole field direction.

stable Eurasia. Yang and Besse [36] compiled Cretaceous paleomagnetic poles from the North China Block (NCB) and calculated the mean pole (latitude=76.4°N, longitude=208.8°E, A_{95}=5.1°). These Cretaceous mean pole positions for stable Eurasia and the NCB are similar to the reference pole position of the stable body of the SCB determined in the present study (Fig. 8-right). This observation indicates that no relative tectonic motion has been detected among stable Eurasia, NCB, and the stable body of the SCB since the Cretaceous. This result leads to the conclusion that the India–Asia collision has not produced relative motion among stable Eurasia, NCB, and the stable body of the SCB, as already documented by Gilder and Courtillot [37]. Even if these tectonic blocks have been affected by the collision, any relative motion among them cannot be detected via paleomagnetic analysis.

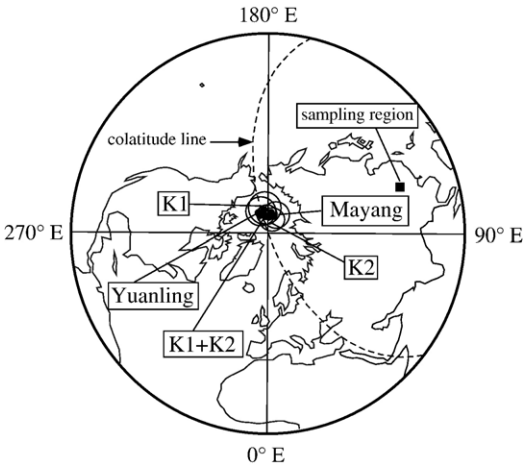


Fig. 7. Geographic (the Yuanling and Mayang Basins), Early Cretaceous (K1), Late Cretaceous (K2), and combined (K1+K2) mean paleomagnetic poles from the Yuanma Basin. The black square indicates the studied region. The colatitude line to the sampling region is also shown.

5. Conclusions

Our paleomagnetic study of Cretaceous red sandstones collected at 30 sites from within the Yuanma Basin in the Hunan Province, central SCB, provided the following conclusions:

(1) On the basis of the direction–correction tilt test [34], the higher temperature components isolated by progressive thermal demagnetization and principal component analysis [32] are judged to be the primary magnetic components acquired prior to the tilting of sedimentary layers.
(2) The poles from the Yuanling and Mayang Basins are indistinguishable from each other at the 95% confidence level. The poles for K1 and K2 are also indistinguishable from each other at the 95% confidence level. The similar positioning of these paleomagnetic poles suggests the absence of Cretaceous and post-Cretaceous tectonic movement in this area.
(3) The Cretaceous (K1+K2) mean pole calculated using the virtual geomagnetic poles of all sites is in agreement with previously reported Cretaceous poles from the eastern [18,19] and western SCB [8]. This agreement in pole locations indicates that the

Table 2
Cretaceous paleomagnetic poles calculated for areas within the stable body of the South China Block

Area (Province)	Location		Age	Pole position			Reference
	Lat (degree N)	Long (degree E)		Lat (degree N)	Long (degree E)	A_{95} (degree)	
Eastern part							
Zhejiang	28.8	120.1	K1	81.8	207.1	6.3	18
Zhejiang	28.8	119.8	K2	81.0	214.3	6.4	18
Fujian	25.9	117.2	K2	79.7	201.7	5.8	18
Guandomg	24.1	115.4	K2	80.8	177.7	8.0	18
Anhui	30.8	118.1	K	74.7	212.5	4.2	19
Shandong	37.0	120.9	K	81.3	217.3	5.9	19
Western part							
Sichuan	28.2	102.5	K	76.9	211.6	6.0	8
Central part							
Hunan	28.2	110.2	K	82.0	178.6	3.9	This study
Mean				80.1	204.1	2.5	

Hunan region has been part of the stable body of the SCB since the Cretaceous. In turn, this conclusion supports the hypothesis [9,18] that a large part of the SCB (excluding a 400-km-wide swath along the RRF) has behaved as a stable body since the Cretaceous.

(4) Cretaceous paleomagnetic poles from the stable body of the SCB are not spread along a small circle centered on the sampling region. Thus, within the SCB, the India–Asia collision has resulted only in local tectonic deformation in the vicinity of the RRF; its influence has not extended to the interior of the stable body of the SCB.

(5) The reference Cretaceous pole position (latitude=80.1°N, longitude=204.1°E, A_{95}=2.5°) of the SCB calculated using eight pole positions from the stable body of the SCB is similar to the Cretaceous mean pole positions from stable Eurasia [22] and the NCB [36]. This result indicates that the India–Asia collision has not caused relative motion among stable Eurasia, NCB, and the stable body of the SCB.

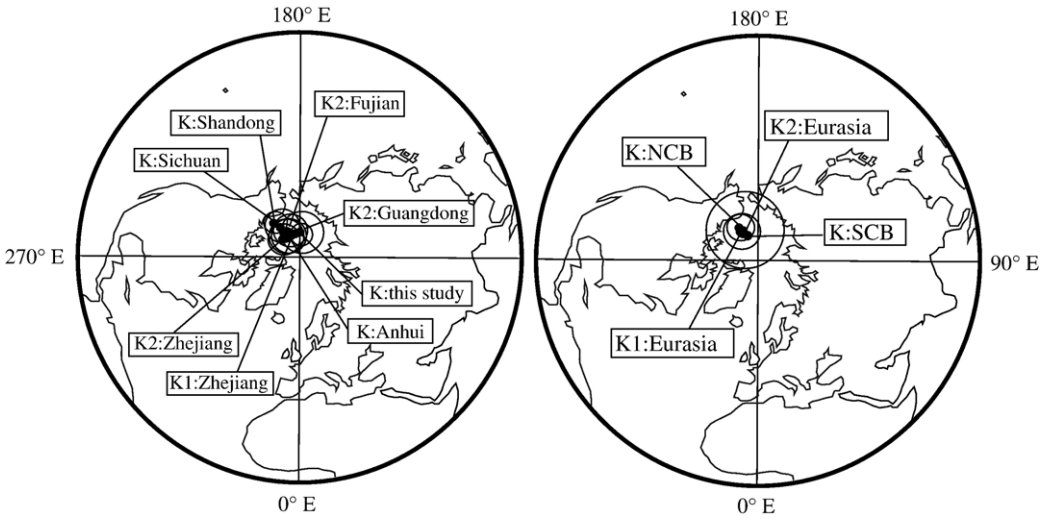


Fig. 8. Left figure: Cretaceous paleomagnetic poles from the South China Block. Data from the eastern SCB comprise K1 and K2 of the Zhejiang Province, K2 of the Fujian Province, K2 of the Guangdong Province, K of the Anhui and Shandong Provinces [18,19]. K of Sichuan is from the western SCB (compiled by Gilder et al. [8]), while K of Hunan Province is from the central SCB, as determined in the present study. Right figure: mean Cretaceous paleomagnetic poles for the stable body of the SCB (K) determined in the present study, stable Eurasia (K1 and K2) [22], and the North China Block (K) [36].

Acknowledgements

We appreciate the informative and critical comments of R. Enkin and S. Gilder. The present study was supported by a project of the National Natural Science Foundation of China (40334037) and a science and technology project of the Ministry of Land and Resources of China (001108) awarded to the co-author (Y. Liu).

References

[1] K. Huang, N. Opdyke, Paleomagnetism of Cretaceous to lower Tertiary rocks from Southwestern Sichuan, a revisit, Earth Planet. Sci. Lett. 112 (1992) 29–40.
[2] Y. Otofuji, Y. Inoue, S. Funahara, M. Murata, X. Zheng, Palaeomagnetic study of eastern Tibet-deformation of the Three Rivers region, Geophys. J. Int. 103 (1990) 85–94.
[3] K. Sato, Y. Liu, Z. Zhu, Z. Yang, Y. Otofuji, Paleomagnetic study of middle Cretaceous rocks from Yunlong, western Yunnan, China: evidence of southward displacement of Indochina, Earth Planet. Sci. Lett. 165 (1999) 1–15.
[4] S. Funahara, N. Nishiwaki, F. Murata, Y. Otofuji, Y. Wang, Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan–Thai–Malay block of western Yunnan, China, Earth Planet. Sci. Lett. 117 (1993) 29–42.
[5] K. Huang, N. Opdyke, Paleomagnetic results from Cretaceous and Jurassic rocks of South and Southwest Yunnan, evidence for large clockwise rotations in the Indochina and Shan–Thai–Malay terranes, Earth Planet. Sci. Lett. 117 (1993) 507–524.
[6] H. Chen, J. Dobson, F. Heller, J. Hao, Paleomagnetic evidence for clockwise rotation of the Shimao region since the Cretaceous: a consequence of India–Asia collision, Earth Planet. Sci. Lett. 134 (1995) 203–217.
[7] K. Takemoto, N. Halim, Y. Otofuji, T.V. Tri, L.V. De, S. Hada, New paleomagnetic constraints on the extrusion of Indochina: Late Cretaceous results from the Song Da terrane, northern Vietnam, Earth Planet. Sci. Lett. 229 (2005) 273–285.
[8] S. Gilder, R. Coe, H. Wu, G. Kuang, X. Zhao, Q. Wu, X. Tang, Cretaceous and Tertiary paleomagnetic results from Southeast China and their tectonic implications, Earth Planet. Sci. Lett. 117 (1993) 637–652.
[9] Y. Liu, H. Morinaga, Cretaceous palaeomagnetic results from Hainan Island in south China supporting the extrusion model of Southeast Asia, Tectonophysics 301 (1999) 133–144.
[10] Y. Otofuji, Y. Liu, M. Yokoyama, M. Tamai, J. Yin, Tectonic deformation of the southwestern part of the Yangtze craton inferred from paleomagnetism, Earth Planet. Sci. Lett. 156 (1998) 47–60.
[11] S. Funahara, N. Nishiwaki, M. Miki, F. Murata, Y. Otofuji, Y. Wang, Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China, implications for the India–Asia collision, Earth Planet. Sci. Lett. 113 (1992) 77–91.
[12] P. Tapponnier, G. Peltzer, A.Y. Le Dain, R. Armijo, P. Cobbold, Propagating extrusion Tectonics in Asia: new insights from simple experiments with plasticine, Geology 10 (1982) 611–616.
[13] P. Tapponnier, G. Peltzer, R. Armijo, On the mechanics of the collision between India and Asia, in: M.P. Coward, A.C. Ries (Eds.), Collision Tectonics, Geol. Soc. Spec. Publ., vol. 19, 1986, pp. 115–157.
[14] G. Peltzer, P. Tapponnier, Formation and evolution of strike–slip faults, rifts, and basins during the India–Asia collision: an experimental approach, J. Geophys. Res. 93 (1988) 15085–15117.
[15] A. Replumaz, P. Tapponnier, Reconstruction of the deformed collision zone between India and Asia by backward motion of lithospheric blocks, J. Geophys. Res. 108 (2003), doi:10.1029/2001JB000661.
[16] The Ministry of Geology and Mineral Resources of China, Geological Map of China, Geological Publishing House, Beijing, 1990 (in Chinese).
[17] R. Zhu, Z. Yang, H. Wu, X. Ma, B. Huang, Z. Meng, D. Fang, Paleomagnetic constraints on the tectonic history of the major blocks of China during the Phanerozoic, Sci. China 41 (1998) 1–19.
[18] H. Morinaga, Y. Liu, Cretaceous paleomagnetism of the eastern South China Block: establishment of the stable body of SCB, Earth Planet. Sci. Lett. 222 (2004) 971–988.
[19] S. Gilder, P.H. Leloup, V. Courtillot, Y. Chen, R.S. Coe, X. Zhao, W. Xiao, N. Halim, J. Cogne, R. Zhu, Tectonic evolution of the Tancheng–Lujiang (Tan–Lu) fault via Middle Triassic to Early Cenozoic paleomagnetic data, J. Geophys. Res. 104 (1999) 15365–15390.
[20] R. Enkin, V. Courtillot, L. Xing, Z. Zhang, Z. Zhuang, J. Zhang, The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block), Tectonics 10 (1991) 547–559.
[21] Z.W. Zhu, T. Hao, H. Zhao, Paleomagnetic study on the tectonic motion of Pan–Xi block and adjacent area during the Yinzhi–Yanshan period, Acta. Geophys. Sin. 31 (1988) 420–431.
[22] R. Enkin, Z. Yang, Y. Chen, V. Courtillot, Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present, J. Geophys. Res. 97 (1992) 13953–13989.
[23] K. Narumoto, Z. Yang, K. Takemoto, H. Zaman, H. Morinaga, Y. Otofuji, Anomalously shallow inclination in middle-northern part of the South China Block: palaeomagnetic study of Late Cretaceous red beds from Yichang area, Geophys. J. Int. 164 (2006) 290–300.
[24] J. Xu, G. Zhu, W. Tong, K. Cui, Q. Liu, Formation and evolution of the Tancheng–Lujiang wrench fault system, a major shear system to the northwest of the Pacific Ocean, Tectonophysics 134 (1987) 273–310.
[25] R. Schlische, Structural and stratigraphic development of the Newark extensional basin, eastern North America, evidence for the growth of the basin and its bounding structures, Geol. Soc. Amer. Bull. 104 (1992) 1246–1263.
[26] Bureau of Geology and Mineral Resources of Zhejiang Province, Rocks and Formations in Zhejiang Province, Publishing House of China University of Geosciences, 1996, 236 pp (in Chinese).
[27] G. Zheng, Cretaceous stratigraphic sequence and structural evolution of Yuanling–Mayang basin, Hunan Geology 17 (1998) 91–95 (in Chinese with English abstract).
[28] Hunan Bureau of Geology and Mineral Resources, Regional Geology of Hunan Province, Geological Publishing House, Beijing, 1988, pp. 557–558, (in Chinese).
[29] Hunan Bureau Of Geology And Mineral Resources, Stratigraphy (Lithostratic) Of Hunan Province, China University Of Geoscience Press, Wuhan, 1997, pp. 251–255, (in Chinese).
[30] G. Zheng, X. Lou, G. Hou, L. Chen, Classification and lithostratigraphy of the Cretaceous system in Yuanling–Mayang basin, J. Stratigraphy 22 (1998) 201–205+215 (in Chinese with English abstract).
[31] M. Mandea, S. Macmillan, International Geomagnetic Reference Field — the eighth generation, Earth Planets Space 52 (2000) 1119–1124.

[32] J.L. Kirschvink, The least-squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[33] J.D.A. Zijderveld, A.C. demagnetization of rocks: analysis of results, in: D.W. Collinson, K.M. Creer, S.K. Runcorn (Eds.), Methods in Palaeomagnetism, Elsevier, Amsterdam, 1967, pp. 254–286.
[34] R. Enkin, The direction–correction tilt test: an all-purpose tilt/fold test for paleomagnetic studies, Earth Planet. Sci. Lett. 212 (2003) 151–166.
[35] J. Besse, V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian Plates, and true polar wander since 200 Ma, J. Geophys. Res. 96 (1991) 4029–4050.
[36] Z. Yang, J. Besse, New Mesozoic apparent polar wander path for south China: tectonic consequences, J. Geophys. Res. 106 (2001) 8493–8520.
[37] S. Gilder, V. Courtillot, Timing of the North–South China collision from new middle to late Mesozoic paleomagnetic data from the North Chine Block, J. Geophys. Res. 102 (1997) 17713–17727.